HIGH-FIDELITY SPIN QUBIT OPERATION AND ALGORITHMIC INITIALIZATION ABOVE 1 K

AUTHORS

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OBJECTIVE

- Demonstrate high-fidelity silicon spin-qubit control above 1 K.
- Achieve deterministic two-qubit algorithmic initialization when $\mathbf{K_BT > hf_{qubit}}$.
- Characterize SPAM fidelities and single and two qubit gate fidelities at elevated temperatures and different magnetic fields.

ABSTRACT

- This poster demonstrates high-fidelity operation of silicon spin qubits above 1 K.
- Using a SiMOS double quantum dot with isotopically enriched silicon, the authors develop an algorithmic initialization protocol based on real-time FPGA control to prepare a pure two-qubit state, even when thermal energy exceeds the qubit energy.
- They achieve readout and initialization fidelities of 99.34%, single-qubit Clifford gate fidelity up to 99.85%, and a two-qubit DCZ gate fidelity of 98.92% at 1 K.

RESULT

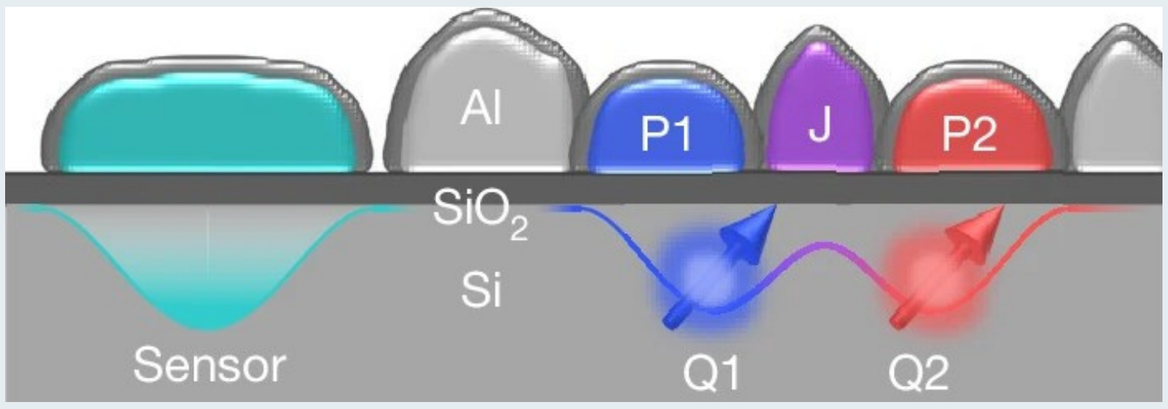
- Achieved 99.34% readout and initialization fidelity at 1 K.
- Demonstrated single-qubit gate fidelity up to 99.85%.
- Achieved two-qubit gate fidelity of 98.92% at 1 K.
- Maintained initialization fidelity above 99% at 85 mT.
- Initialized qubits in approximately 150 μs, averaging three iterations.
- Demonstrated high-fidelity quantum operation even when thermal energy exceeds qubit energy.

INTRODUCTION

- Silicon spin qubits are promising candidates for scalable quantum computing because they offer low error rates, long information-holding times, and compatibility with established semiconductor manufacturing technologies.
- Large-scale quantum processors require many densely packed qubits. At millikelvin temperatures, the available cooling power of cryogenic systems becomes a major limitation as the number of qubits increases.
- Operating above 1 K provides substantially higher cooling power and could improve the scalability of quantum processors. However, at these temperatures, the thermal energy can become comparable to or greater than the qubit energy:

$$K_BT \gtrsim hf_{qubit}$$

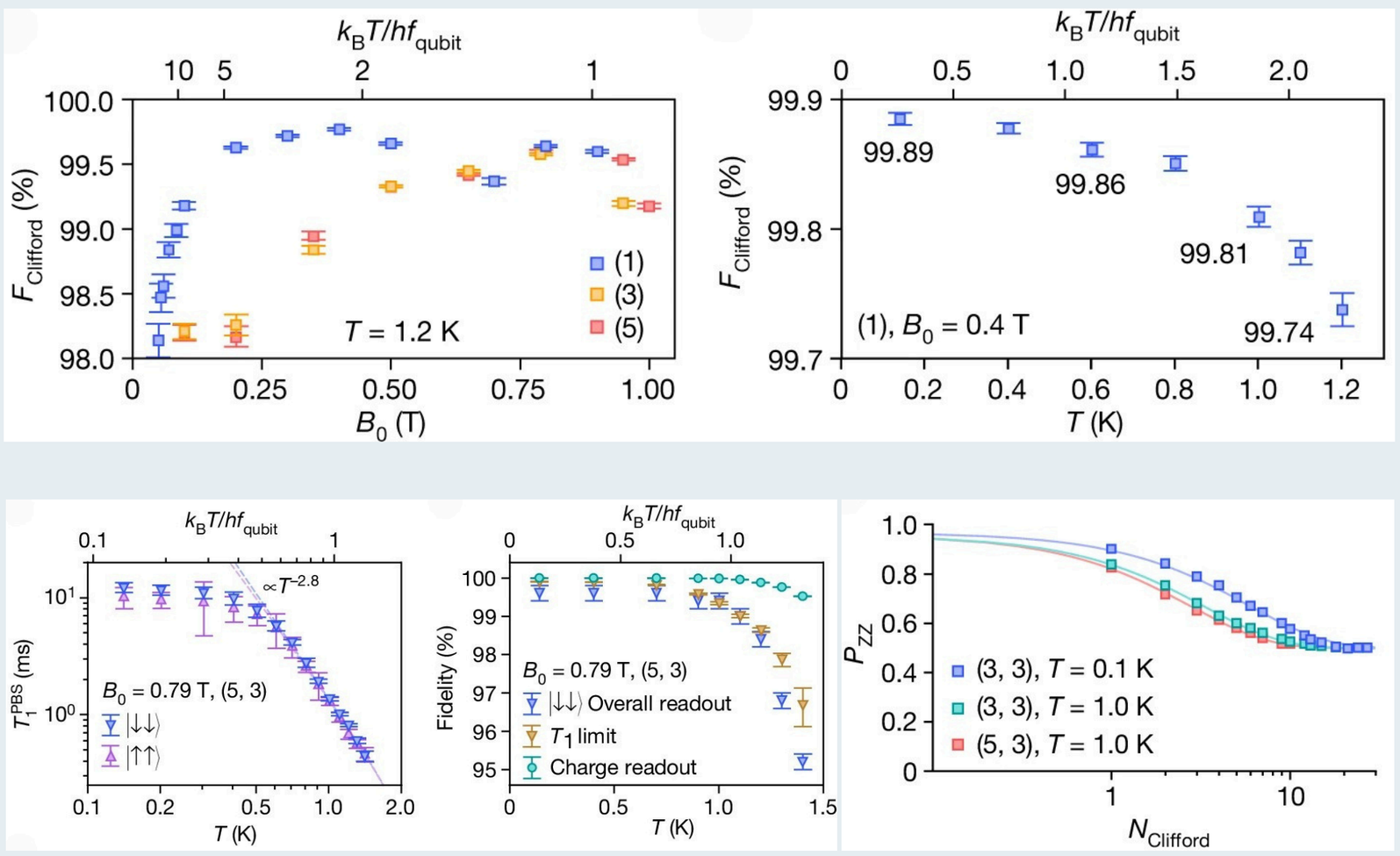
- Under these conditions, qubits become thermally mixed, making conventional initialization and high-fidelity operation more difficult. Previous demonstrations of operation above 1 K showed feasibility but experienced degradation in state-preparation-and-measurement (SPAM) and gate fidelities.
- This work addresses the thermal initialization challenge using algorithmic initialization, combining two-qubit logic, parity-based single-shot readout, and feedback to transfer entropy from the qubits to the environment.

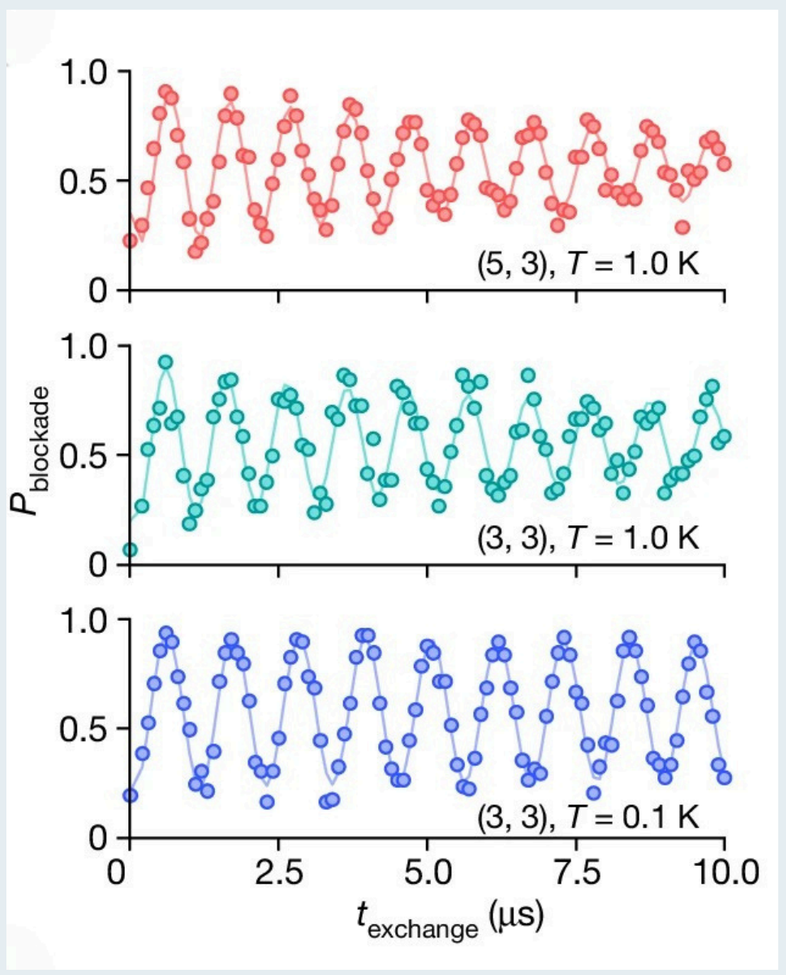


DEVICE CROSS-SECTION: SPIN QUBITS Q1, Q2 AND RF-SET SENSOR

ERROR SOURCES AND LIMITATIONS

- Exchange interactions and microwave noise cause gate errors.
- Thermal relaxation affects qubit stability.
- Charge-sensor noise reduces readout accuracy.
- Initialization takes approximately 100–200 μs.
- Higher temperatures may reduce performance.
- Further improvements are needed for fault-tolerant computing.





REFERENCES

1. J. Y. Huang et al., "High-fidelity spin qubit operation and algorithmic initialization above 1 K," Nature, vol. 627, no. 8005, pp. 772–777, Mar. 2024, doi: 10.1038/s41586-024-07160-2.
2. C. H. Yang et al., "Operation of a silicon quantum processor unit cell above one kelvin," Nature, vol. 580, pp. 350–354, 2020, doi: 10.1038/s41586-020-2171-6.
3. L. Petit et al., "Universal quantum logic in hot silicon qubits," Nature, vol. 580, pp. 355–359, 2020, doi: 10.1038/s41586-020-2170-7.
4. M. A. I. Johnson et al., "Beating the thermal limit of qubit initialization with a Bayesian Maxwell's demon," Physical Review X, vol. 12, Art. no. 041008, 2022, doi: 10.1103/PhysRevX.12.041008.